Thinking Profile

Thinking Profile helps you plan a whole unit around one kind of thinking, so students practise it in four to six tasks and can see for themselves that they are getting better. Pick one kind of thinking for the unit, plan four to six tasks around it, and let students rate every answer on the same four-step ladder.

One cluster per unit

Generating ideas
Producing many and unusual options.

Evaluating and choosing
Weighing evidence, testing claims and deciding.

Comparing and transferring
Seeing the structure one case shares with another.

Explaining and reasoning from basics
Finding the best explanation and rebuilding from what is certainly true.

Causes and systems
Chains, loops and side effects.

Regulating your own thinking
Watching and adjusting how you think. It makes every other cluster sharper.

After every task, students ask

1. Which step did my answer reach?
2. What in my answer shows it?
3. My next step

The ladder is for students' own self-check. It is never graded.

The ladders are Choosewise's own, built on the logic of the SOLO taxonomy. They have not been tested in research.

What the labels mean

Strong Meta-analyses of classroom interventions show that teaching this kind of thinking improves it.

Promising Controlled studies show gains, but results are mixed, small in number or dependent on context.

Framework A useful structure for thinking, with no intervention studies behind it yet.

CLUSTER 1

Generating ideas

Producing many and unusual options.

1 I write down one idea that works.

Shows it: Point to the idea and to what makes it work.

2 I write down several ideas that work in different ways, not small changes of the same idea.

Shows it: Point to each idea and to what makes it different from the others.

3 I sort my ideas into kinds and add one from a kind I had missed.

Shows it: Point to your groups, their names and the idea you added to fill the gap.

4 I pick the idea that fits best when one condition changes, explain why, and say how sure I am.

Shows it: Point to the changed condition, the idea you picked, your reason and the words that say how sure you are.

IN CLASS

Students solve one problem in as many different ways as they can, then hunt for the most unusual way.

TRAINED BY

- Many ways Promising

EVIDENCE

Strong Teaching students to plan, monitor and evaluate their own learning (metacognition).

Promising Self-assessment against concrete criteria.

Framework These ladders, built on the logic of the SOLO taxonomy and not tested in research.

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CLUSTER 2

Evaluating and choosing

Weighing evidence, testing claims and deciding.

1 I make a choice or a judgement and give one reason for it.

Shows it: Point to your choice and the one reason.

2 I give several reasons, each backed by a fact or a rule I can check.

Shows it: Point to each reason and the fact or rule beside it.

3 I weigh my reasons against each other and against the other side, and show which one decides it.

Shows it: Point to where you answer the other side and the sentence that says which reason weighs most.

4 I test my judgement on a new case or a new piece of evidence, and say how sure I am.

Shows it: Point to the new case or evidence, whether your judgement held, and the words that say how sure you are.

IN CLASS

Students set criteria or say what evidence would convince them before they decide, then defend their choice against the other side.

TRAINED BY

- Pick and justify Promising
- Check the claim Strong
- Argue it Promising
- Flawed Answer

EVIDENCE

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Promising Self-assessment against concrete criteria.

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CLUSTER 3

Comparing and transferring

Seeing the structure one case shares with another.

1 I name one thing the two cases share.

Shows it: Point to the shared feature in both cases.

2 I name several things that match and at least one place where the cases differ.

Shows it: Point to each match and to the difference.

3 I explain the rule that makes the cases match, and where the comparison breaks.

Shows it: Point to the sentence that states the rule and to the place where the comparison breaks.

4 I use the rule on a new case, and say how sure I am that it fits.

Shows it: Point to the new case, your rule used on it, and the words that say how sure you are.

IN CLASS

Students set two cases side by side, map what matches, write the rule the comparison shows and try it on a new case.

TRAINED BY

- Like what? Strong
- Side by Side

EVIDENCE

Strong Teaching students to plan, monitor and evaluate their own learning (metacognition).

Promising Self-assessment against concrete criteria.

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CLUSTER 4

Explaining and reasoning from basics

Finding the best explanation and rebuilding from what is certainly true.

1 I give one explanation or one fact I know for certain.

Shows it: Point to the explanation or the fact.

2 I give several explanations or steps, and say what each one rests on.

Shows it: Point to each explanation or step and what it rests on: evidence, a known fact or an assumption.

3 I link my steps into one line of reasoning, or show which explanation fits the evidence best.

Shows it: Point to the linking words (because, so, which means) or to the evidence that rules out the other explanations.

4 I use my reasoning on a new case or a changed number, say where it stops working, and say how sure I am.

Shows it: Point to the new case, the place where your reasoning stops working, and the words that say how sure you are.

IN CLASS

Students explain a puzzling result, rebuild a rule from what they know for certain, or estimate and model a real question, saying every assumption out loud.

TRAINED BY

- Best explanation Promising
- Strip it down Framework
- Estimate it Framework
- Model it Promising

EVIDENCE

Strong Teaching students to plan, monitor and evaluate their own learning (metacognition).

Promising Self-assessment against concrete criteria.

Framework These ladders, built on the logic of the SOLO taxonomy and not tested in research.

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CLUSTER 5

Causes and systems

Chains, loops and side effects.

1 I name one cause and one effect.

Shows it: Point to the cause, the effect and the word that links them.

2 I build a chain: A leads to B leads to C.

Shows it: Point to each link in the chain: at least three parts in a row.

3 I find a loop: C affects A again, or a delay or a side effect.

Shows it: Point to the arrow or sentence that leads back to the start, or that names the delay or side effect.

4 I use the loop to predict what happens if one part changes, and say how sure I am.

Shows it: Point to the part you changed, your prediction and the words that say how sure you are.

IN CLASS

Students follow a decision through its consequences, or draw how the parts of a system affect each other and find the loops.

TRAINED BY

- Then what? Framework
- Map the loop Promising

EVIDENCE

Strong Teaching students to plan, monitor and evaluate their own learning (metacognition).

Promising Self-assessment against concrete criteria.

Framework These ladders, built on the logic of the SOLO taxonomy and not tested in research.

The ladder is for students' own self-check. It is never graded.

CLUSTER 6

Regulating your own thinking

Watching and adjusting how you think. It makes every other cluster sharper.

1 I name one thing I did while I worked.

Shows it: Point to the strategy you name, such as "I drew it". "I thought harder" does not count.

2 I describe several steps in how I worked, and one moment that shaped how I worked, such as changing my approach or getting stuck.

Shows it: Point to each step and to the moment that shaped how you worked.

3 I explain what made me change my approach or moved me on, and why it helped.

Shows it: Point to the sentence that says what made you change your approach or moved you on, and why it helped.

4 I say what I would change next time and why, and how sure I am that it would help.

Shows it: Point to what you would change, your reason and the words that say how sure you are.

IN CLASS

At the end of any task, students write for two minutes about how they thought, not what they found, and what they would do differently.

TRAINED BY

- Think back Strong

EVIDENCE

Strong Teaching students to plan, monitor and evaluate their own learning (metacognition).

Promising Self-assessment against concrete criteria.

Framework These ladders, built on the logic of the SOLO taxonomy and not tested in research.

The ladder is for students' own self-check. It is never graded.

WORKED EXAMPLE

Biology, Grade 8: Ecosystems

6 weeks · following Causes and systems

THE THINKING THREAD

One question runs through the unit: when one thing in an ecosystem changes, what else changes, and when? Each week students follow the effects a little further: a chain in week 1, a loop in week 2, a delayed side effect in week 3, and by week 6 a prediction they back with the loop.

WEEK	TASK	WHAT STUDENTS DO	ROOM FOR
Week 1	Then what?	Queensland, 1935: cane toads are released to eat the beetles that damage sugar cane. Students write the first consequence, the consequence of that, and one more.	Step 2
Week 2	Map the loop	Students draw how snowshoe hares, lynx and the shrubs the hares eat affect each other, find one loop and say whether it grows or balances itself.	Step 3
Week 3	Then what?	Harder inputs: a farmer spreads extra fertiliser next to a lake. Students follow the chain into the water and find the delayed side effect: algae bloom, die and rot, and the fish run short of oxygen.	Step 3
Week 4	Best explanation	Borrowed from Explaining. 29 reindeer released on St. Matthew Island in 1944 grew to about 6,000 by 1963, then fell to 42 by 1966. Students propose explanations and say what evidence would tell them apart. Researchers still disagree: overgrazed lichen plus a hard winter, or the winter alone.	Step 3
Week 5	Side by Side: One thing changed	Borrowed from Comparing. Two identical forests have too many deer, and both remove the same number every year. In A most of the deer removed are females, in B most are males. Students compare what happens, write the rule and predict a third forest.	Step 4
Week 6	Map the loop	Back to the hares, now with less support: a disease halves the lynx. Students use their loop to predict what happens to hares and shrubs over the next few years, and say how sure they are.	Step 4

TELL STUDENTS

For six weeks we practise one kind of thinking: causes and systems. After every task you check your answer against the same four-step ladder, point to the words that show your step, and write your next step. Nobody grades the ladder. It is there so you can see yourself climb.

WITHOUT THE WEBSITE

Your own unit plan

Replace everything in {braces}, then type the prompt into any AI tool: Copilot, Gemini, ChatGPT or Claude. Check the plan before the unit, especially anything under CHECK.

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Do not include student names or student work in this conversation.

You are helping a Grade {grade} {subject} teacher plan a {weeks}-week unit so that students practise one kind
of thinking all the way through it, instead of handing their thinking to AI. Students do the tasks on paper,
without AI. After each task they mark which step of a four-step ladder their answer reached. The ladder is for
students' own self-check. It is never graded.

Unit topic: {topic}
Unit outline: {overview}   (optional)

Thinking to follow: {cluster name}. {cluster description}
The ladder:
1. {step 1}
2. {step 2}
3. {step 3}
4. {step 4}

Each task below is made in its own free builder on choosewise.education; the teacher types in the inputs you
give, and the builder makes the task sheet students answer on paper.
Builder fields. All three: Subject, Grade, What is the lesson about? (at most 400 characters), What do
students already know? (optional). Thinking Moves also has Thinking move (the task name); grades 8 to 12.
Flawed Answer also has Flaw type and How hard should the flaw be to spot? (Easier to spot or Harder to spot);
grades 8 to 12. Side by Side also has Pair type and How hard should the difference be to see? (Easier to see
or Harder to see); grades 7 to 12. If the unit grade is not offered, use the nearest. The builder writes the
task itself, so put the case, numbers or text for students into What is the lesson about?
Preferred tasks, because they train {cluster name}:
List every task from the Trained by boxes on pages 2–7, preferred ones first.

Rules: At least half of the tasks in the plan come from the followed cluster's preferred tasks. A task from
another cluster may be borrowed only when its content makes students use the followed cluster's thinking.
Repeating the same task is intended when each round has harder inputs: more variables, less familiar content
or less support. Demand rises across the unit, so the later tasks give room to reach steps 3 and 4. Students
rate every task on the followed cluster's ladder, whatever tool the task came from. Do not invent studies or
sources.

Reply in plain text, no tables, in this order:
THE THINKING THREAD: one or two sentences on what kind of thinking this unit trains and why it fits the topic
PLAN: four to six tasks spread across the weeks. For each task write
WEEK:
TASK: the task name and the tool
WHAT STUDENTS DO: one sentence
ROOM FOR STEP: the highest step this task gives room for
BUILDER INPUTS: one Field: value line per field listed above; skip a field you leave empty
LINK: the builder link
TELL STUDENTS: three or four sentences the teacher reads at the start of the unit, introducing the ladder and
saying it is never graded
CHECK: facts or claims in the plan that could be wrong, one per line, at most four lines. Write "None" if
there are none.
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Regulating your own thinking. For this cluster use these rules instead of the Rules line above: Any task from any tool may be used, because every task ends with a Think back step. Repeating the same task is intended when each round has harder inputs: more variables, less familiar content or less support. Demand rises across the unit, so the later tasks give room to reach steps 3 and 4. Students rate the ladder on the Think back part at the end of each task, not on the task answer.

Faster online: the planner at choosewise.education/thinking-profile fills this in for you and picks the preferred tasks. Students need no AI for the tasks. If older students use AI themselves, check the tool's age rules, your national rules and your school policy first.